VLM Reliability Toolkit: Unified Inference-Time Hallucination Detection and Attribute-Binding Correction for Vision-Language Models

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***Abstract*—Vision-language models (VLMs) describe images fluently, yet remain unreliable in two documented ways: they hallucinate objects that are absent from the image, and they mis-bind attributes to the wrong objects (e.g., a "red car and blue shirt" when the colours are swapped). These failures have been studied separately: hallucination work verifies whether an object exists, while compositionality work diagnoses binding failures without correcting them. This paper presents the *VLM Reliability Toolkit*, a training-free, model-agnostic, inference-time framework that decomposes a VLM's output into atomic claims, verifies existence claims with open-vocabulary grounding and attribute-binding claims with region-level attribute classification, and then rewrites, re-ranks, or flags unsupported content while leaving verified content untouched. A reliability score that fuses claim-level verification, model confidence, and inter-candidate agreement drives the decision policy. We describe the architecture, scoring and decision rules, and a reproducible evaluation protocol spanning POPE, CHAIR, Winoground, and SugarCrepe, with explicit targets for accuracy gain and latency overhead. The framework is intended as a drop-in reliability layer for existing VLMs, with no retraining of the base model.**

***Index Terms*—vision-language models, object hallucination, attribute binding, compositionality, post-hoc correction, claim verification, inference-time reliability.**

I. Introduction

Vision-language models (VLMs) such as LLaVA [1] and BLIP-2 [2] have moved quickly from research prototypes to deployed systems for visual search, content moderation, and accessibility tooling. Despite their fluency, their outputs cannot yet be trusted at the level of individual statements. The first well-documented failure is *object hallucination*, where the model mentions objects, entities, or events that are not in the image [3]–[5]. It is largely attributed to language-prior bias and object co-occurrence statistics learned during training, rather than to grounded visual evidence [4], [5].

The second failure is the *attribute-binding error*: the model perceives the correct objects and attributes but assigns an attribute to the wrong object, for example "a blue cup on a red table" for an image of a red cup on a blue table [6]–[8]. Nothing is invented; the relationship between correctly perceived elements is wrong. This failure is structurally distinct from hallucination and requires a different verification strategy: confirming that an object exists is not the same as confirming that a given attribute belongs to it.

Most remedies operate at training time through supervised fine-tuning, preference optimisation, or contrastive decoding. These are costly and must be repeated for every new model checkpoint; aligning a frontier-scale model has reportedly cost tens of millions of dollars [9]. Post-hoc, inference-time correction offers an alternative that leaves model weights untouched [10]–[12], but existing methods concentrate on object-level hallucination and do not audit attribute binding in the same pass.

This paper proposes a unified framework that closes this gap. Its contributions are:

1) A single claim-based diagnostic that represents *both* object-existence and attribute-binding statements in a common form, so that one pipeline detects and corrects both failure modes.
2) A reliability score and decision policy (accept, correct, re-rank, abstain) that is calibrated against model confidence and inter-candidate agreement.
3) An evaluation protocol that places POPE, CHAIR, Winoground, and SugarCrepe in one repeatable harness, including an adaptation for the image–text matching benchmarks.
4) A model-agnostic implementation plan, including a dashboard with provenance logging, for use as a drop-in layer over any HuggingFace-compatible VLM.

This paper describes the design and evaluation plan of the toolkit. Empirical results will be reported once the evaluation described in Section V has been completed; no performance figures for the proposed system are claimed here.

II. Related Work

A. Hallucination Evaluation and Mitigation

CHAIR [3] formalised object hallucination by comparing the objects named in a caption with ground-truth annotations. POPE [4] recasts the problem as polling: the model answers yes/no questions about object existence under random, popular, and adversarial negative sampling, and exposes how strongly VLMs hallucinate objects that frequently co-occur with real ones. HallusionBench [14] separates language-driven hallucination from visual illusion using 346 images and 1,129 control-paired questions, and a survey [5] organises the broader literature.

Mitigation methods fall into training-based and post-hoc families. LURE [11] identifies co-occurrence, decoding uncertainty, and token position as hallucination determinants

and trains a lightweight "revisor" to rewrite captions; it needs a hallucination-annotated training set and edits objects only. VOLCANO [12] uses iterative self-feedback to revise answers without an external verifier. OPERA [13] intervenes in decoding through an over-trust penalty and retrospection-allocation. Woodpecker [10] is the closest antecedent of our work: a training-free, five-stage pipeline (key-concept extraction, question formulation, visual knowledge validation with external perception models, visual claim generation, and correction) that raised POPE accuracy from 54.67% to 85.33% for MiniGPT-4 and from 62% to 86.33% for mPLUG-Owl, i.e., absolute gains of 30.66 and 24.33 points. Its validation is organised around object-level grounding and does not explicitly test whether an attribute is bound to the correct instance.

B. Compositionality and Attribute Binding

ARO [6] contains over 50,000 test cases built from Visual Genome and COCO/Flickr30k and shows that VLMs often behave like bags-of-words: they recognise objects and attributes in isolation but fail to bind them. Winoground [7] and SugarCrepe [8] probe the same weakness with minimally different caption pairs and, in SugarCrepe, with debiased hard negatives. Lewis *et al.* [15] similarly probe whether CLIP binds concepts. These works are diagnostic: they measure binding failure but do not correct it in generated text.

C. Positioning

Table I summarises the gap. Hallucination methods correct object existence but not binding; compositionality benchmarks measure binding but supply no correction. To our knowledge, few if any existing systems verify both claim types and apply the corresponding corrections within a single inference-time pass.

TABLE I
COMPARISON WITH REPRESENTATIVE METHODS

Method	Training-free	Ext. verifier	Object claims	Attr. binding	Unified
LURE [11]	No	No	Yes	No	No
VOLCANO [12]	Partial	No	Yes	No	No
OPERA [13]	Yes	No	Yes	No	No
Woodpecker [10]	Yes	Yes	Yes	Partial	No
ARO / Winoground / SugarCrepe [6]–[8]	N/A	N/A	No	Diagnosis only	No
Proposed toolkit	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

III. PROBLEM FORMULATION

Let I denote an input image, q a text query, and M a VLM that produces a response $y = M(I, q)$. A response is decomposed into a set of atomic claims $C(y) = \{c_1, \ldots, c_n\}$. Each claim is of one of two types:

- an *existence claim* $\langle o \rangle$, asserting that object o is present;
- a *binding claim* $\langle o, a \rangle$, asserting that attribute a (colour, material, count, or a spatial relation) holds for object o.

Let $O(I)$ be the set of objects truly present and $B(I)$ the set of true object–attribute bindings. An existence claim $\langle o \rangle$ is a **hallucination** if $o \notin O(I)$. A binding claim $\langle o, a \rangle$ is a **binding error** if $o \in O(I)$ and a is a true attribute of some other object in I, but $\langle o, a \rangle \notin B(I)$. The goal is to produce a corrected response $\hat{y}$ in which every claim is either supported by I or removed or flagged, while preserving the informativeness of the original response and without modifying the weights of M.

IV. PROPOSED METHODOLOGY

A. System Architecture

The toolkit is a pipeline of seven cooperating components (Fig. 1): a *VLM Execution Engine* that loads the base model; an *Inference Orchestrator* that requests *n*-best candidates; a *Diagnostic Engine* that extracts and verifies claims; a *Reliability & Scoring Engine*; a *Correction / Decision Layer*; an *Evaluation & Benchmarking* module; and a *Dashboard & Reporting* module for human review. The system runs entirely at inference time and treats the VLM as a black box (or white box when logits are available).

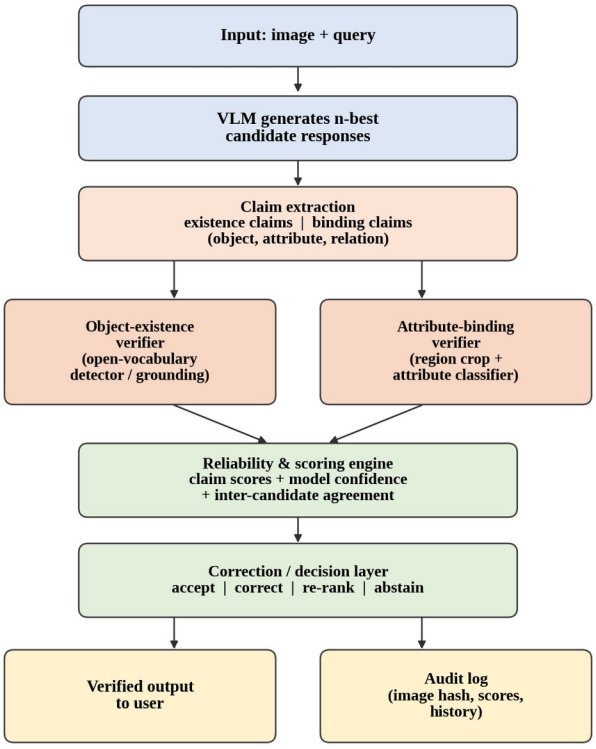


Fig. 1. Inference-time diagnostic and correction pipeline of the VLM Reliability Toolkit.

B. Candidate Generation and Claim Extraction

Rather than a single greedy response, the orchestrator requests n candidate responses $\{y_1, \ldots, y_n\}$. The n-best list serves two purposes: it supplies alternatives for re-ranking, and it provides a signal of agreement across candidates. Each candidate is parsed into existence and binding claims using a rule-and-LLM-based extractor that emits structured tuples (object, attribute type, attribute value, relation target).

C. Claim Verification

Existence claims. An open-vocabulary detector or grounding model (e.g., Grounding DINO [16]) is queried with

the object phrase. The existence score is the highest detection confidence d_b over returned boxes B_o:

$$s^{obj}(o) = \max_{b \in B_o} d_b \quad (1)$$

Binding claims. For $\langle o, a \rangle$, the detector first localises every instance of o. Each box is cropped and passed to an attribute classifier (e.g., region-level image–text alignment or a region-conditioned VQA probe), which returns $P(a \mid \text{crop}_b)$. The binding is supported only if the attribute holds on the *same region* as the object:

$$s^{attr}(o, a) = \max_{b \in B_o} d_b \cdot P(a \mid \text{crop}_b) \quad (2)$$

Because the attribute is evaluated on the localised region of o rather than on the whole image, a colour that exists elsewhere in the scene cannot spuriously validate a mis-bound claim. Spatial-relation claims are checked from the geometry of the detected boxes.

D. Reliability Scoring

Let v_i be the verification score of claim c_i (s^{obj} or s^{attr}). The reliability of candidate y_j combines three signals:

$$R_j = \lambda_1 \cdot \text{mean}_i(v_i) + \lambda_2 \cdot p_j + \lambda_3 \cdot A_j \quad (3)$$

where p_j is the length-normalised mean token probability of the candidate (the VLM's native confidence), A_j is the fraction of its claims also asserted by the other $n-1$ candidates, and $\lambda_1+\lambda_2+\lambda_3 = 1$. The weights are tuned on a held-out split and the resulting score is checked for calibration (Section V).

E. Decision and Correction Policy

Per claim, with configurable thresholds $\tau_{lo} < \tau_{hi}$, the Correction / Decision Layer applies:

1) **Accept** if $v_i \geq \tau_{hi}$; the claim is left untouched.
2) **Correct** if $v_i \leq \tau_{lo}$ and the verifier supplies positive evidence. For a binding error, the attribute is replaced by the highest-probability attribute observed on the region (or swapped with the correctly bound counterpart). For a hallucinated object with no detections, the claim is deleted rather than rewritten, since no evidence supports a substitute.
3) **Re-rank** if the top candidate's reliability is below threshold but another candidate satisfies it, i.e., select $j^* = \text{argmax}_j R_j$.
4) **Abstain / flag** if the score lies between τ_{lo} and τ_{hi}, or if no candidate is reliable; the case is escalated for human review.

Every interaction is written to an audit log (image hash, candidate history, claim scores, action taken), enabling regression testing and provenance review. At the data-flow level, the image and query are cached in an image/query store, passed through candidate generation, claim extraction and verification, and scoring/decision, and the corrected output is returned to the user while the log entry is written (Fig. 1).

V. Evaluation Protocol

A. Benchmarks and Metrics

Table II lists the benchmarks, the failure mode each targets, and the metrics reported. Each benchmark split will cover at least 500 images, consistent with standard POPE/CHAIR practice [4].

TABLE II
Benchmarks, Failure Modes, and Metrics

Benchmark	Failure mode	Metrics
POPE [4]	Object existence	Accuracy, precision, recall, F1, yes-ratio on random / popular / adversarial splits
CHAIR [3]	Object hallucination in captions	CHAIR_S (sentence-level), CHAIR_I (instance-level); object recall to measure over-deletion
Winoground [7]	Attribute / relation binding	Text, image, and group scores
SugarCrepe [8]	Attribute / relation binding	Accuracy on replace, swap, and add categories
System-level	Cost and trust	Latency ratio, abstention rate, calibration error of R_j

B. Adapting Matching Benchmarks

POPE and CHAIR evaluate generated or yes/no answers and map directly onto the pipeline. Winoground and SugarCrepe, however, are image–text matching tasks: a model must prefer the true caption over a minimally edited foil. For these, the toolkit scores each candidate caption by decomposing it into claims, verifying them against the image, and using the caption-level reliability R as the matching score. Gains are attributed to the verification layer only by comparing against the same base VLM's uncorrected score.

C. Baselines, Ablations, and Targets

The primary baseline is the uncorrected base VLM (LLaVA and BLIP-2). Ablations remove, in turn, the attribute-binding verifier (an object-only, Woodpecker-style variant), the scoring engine, and n-best generation ($n = 1$). Planned targets are summarised in Table III; the POPE target is set conservatively relative to the 24–31 point gains reported for Woodpecker [10].

TABLE III
Planned Success Targets

Objective	Target
POPE accuracy	≥ 15–20% improvement over the uncorrected baseline
CHAIR	Measurable reduction in CHAIR_S and CHAIR_I, with object recall reported alongside
Winoground / SugarCrepe	Improvement over the base VLM's uncorrected score on each metric
Latency	Under 2× the single-pass baseline inference time
Coverage	≥ 500 images per benchmark split

D. Implementation

The system is implemented in Python 3.10+ with PyTorch and HuggingFace Transformers, using open-source base VLMs and open-vocabulary detectors. A GPU with at least 16 GB VRAM (e.g., NVIDIA T4 or RTX 30-series or higher) is recommended for 7B-parameter VLMs. The pipeline is served through FastAPI, and a Streamlit or Next.js dashboard presents original-versus-corrected outputs, per-claim scores, and flagged cases for quick annotation.

VI. Illustrative Example

Consider an image of a person on a beach wearing a red shirt and blue jeans, with the query "Describe this image in detail." The VLM produces: "A person in a blue shirt and red jeans stands on the beach holding a surfboard." Table IV traces the claims through verification and correction.

TABLE IV
Worked Example of Claim Verification and Correction

Claim	Verification result	Action
person exists	Confirmed	Accept
shirt is blue	Shirt region is red	Correct → red
jeans are red	Jeans region is blue	Correct → blue
surfboard exists	No detection in image	Delete (no evidence)
location is beach	Confirmed	Accept

The corrected output is "A person in a red shirt and blue jeans stands on the beach." The two attribute errors are repaired in place by swapping the bound colours, whereas the hallucinated object is removed, since there is no visual evidence to support a substitute. Verified content is left intact.

VII. Anticipated Outcomes and Limitations

We anticipate a measurable reduction in CHAIR hallucination rates, better detection of adversarial-split hallucinations on POPE without losing accuracy on the other splits, and improved compositional scores on Winoground and SugarCrepe through genuine repair of bindings rather than merely deleting uncertain content. Beyond raw numbers, the study should demonstrate that reliability gains are attainable at inference time without retraining, that the modular pipeline transfers to newer VLMs with minimal reconfiguration, and that the released artefacts (wrapper library, dashboard, and benchmark report) are reusable.

The approach has several limitations. (i) Correction quality is bounded by the verifier: small or occluded objects may be missed, leaving hallucinations uncorrected, and rare or highly specialised categories may fall outside the detector's vocabulary. (ii) Claim extraction errors propagate downstream. (iii) Attributes such as colour are sensitive to lighting and ambiguous boundaries. (iv) Aggressive deletion can raise precision at the cost of informativeness, so recall and object coverage are reported with every hallucination metric. (v) The reliability score must be calibrated iteratively with human-annotated validation data before it can serve as a production confidence measure. (vi) Post-generation correction adds latency, which latency-critical applications may need to trade against correction depth. Reasoning errors, toxicity, and adversarial robustness lie outside the present scope.

VIII. Conclusion

We presented the VLM Reliability Toolkit, a training-free, model-agnostic framework that detects and corrects both object hallucinations and attribute-binding errors in a single inference-time pass. By representing both failure modes as verifiable claims, scoring them jointly, and applying a transparent accept/correct/re-rank/abstain policy, the framework aims to make reliability a property that can be maintained on top of a model rather than embedded in it. Future work will report results on POPE, CHAIR, Winoground, and SugarCrepe, extend verification to richer relational and counting claims, and evaluate real-world deployment and monitoring.

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References

[1] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023.

[2] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023.

[3] A. Rohrbach, L. A. Hendricks, K. Burns, T. Darrell, and K. Saenko, "Object hallucination in image captioning," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2018.

[4] Y. Li, Y. Du, K. Zhou, J. Wang, W. X. Zhao, and J.-R. Wen, "Evaluating object hallucination in large vision-language models," in *Proc. EMNLP*, 2023.

[5] H. Liu *et al.*, "A survey on hallucination in large vision-language models," *arXiv:2402.00253*, 2024.

[6] M. Yuksekgonul, F. Bianchi, P. Kalluri, D. Jurafsky, and J. Zou, "When and why vision-language models behave like bags-of-words, and what to do about it?" in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[7] T. Thrush *et al.*, "Winoground: Probing vision and language models for visio-linguistic compositionality," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022.

[8] C.-Y. Hsieh *et al.*, "SugarCrepe: Fixing hackable benchmarks for vision-language compositionality," in *Proc. NeurIPS*, 2023.

[9] Galileo AI, "How much does LLM training cost?," 2026. [Online]. Available: galileo.ai/blog/llm-model-training-cost

[10] S. Yin *et al.*, "Woodpecker: Hallucination correction for multimodal large language models," *arXiv:2310.16045*, 2023.

[11] Y. Zhou *et al.*, "Analyzing and mitigating object hallucination in large vision-language models," *arXiv:2310.00754*, 2023.

[12] S. Lee *et al.*, "VOLCANO: Mitigating multimodal hallucination through self-feedback guided revision," in *Proc. NAACL*, 2024.

[13] Q. Huang *et al.*, "OPERA: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation," in *Proc. CVPR*, 2024.

[14] T. Guan *et al.*, "HallusionBench: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models," in *Proc. CVPR*, 2024.

[15] M. Lewis, Q. Yu, J. Merullo, and E. Pavlick, "Does CLIP bind concepts? Probing compositionality in large image models," *arXiv:2212.10537*, 2022.

[16] S. Liu *et al.*, "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," *arXiv:2303.05499*, 2023.